19. Given that $K_{sp} = 1 \times 10^{-12}$

$$\underset{s}{Ag_2CrO_4}(s) \rightleftharpoons \underset{2s}{2Ag^+_{(aq)}} + \underset{s}{CrO_4^{-2}}(aq)$$

$$\underset{0.01M}{AgNO_3}(s) \rightleftharpoons \underset{0.01M}{Ag^+_{(aq)}} + \underset{0.01M}{NO_3^-}(aq)$$

$$K_{sp} = [Ag^+]^2\,[CrO_4^{2-}]$$

$$\begin{bmatrix} [Ag^+] = 2s + 0.01 \\ 0.01 >> 2S \\ \therefore\ [Ag^+] = 0.01M \\ [CrO_4^{2-}] = s \end{bmatrix}$$

$$1 \times 10^{-12} = (0.01)^2\ (s)$$

$$(s) = \frac{1 \times 10^{-12}}{(10^{-2})^2} = 1 \times 10^{-8}\,M$$

20. $\underset{s}{Ca_3(PO_4)_2} \rightleftharpoons \underset{3s}{3Ca^{2+}} + \underset{2s}{2PO_4^{3-}}$

$K_{sp} = [Ca^{2+}]^3[PO_4^{3-}]^2$

$K_{sp} = (3s)^3(2s)^2$

$K_{sp} = 27s^3.4s^2$

$K_{sp} = 108s^5$

21. $CaF_2(s) \rightleftharpoons Ca^{2+}_{(aq)} + 2F^-(aq)$

$[F^-] = 2\,[Ca^{2+}] = 2 \times 3.3 \times 10^{-4}M$

$= 6.6 \times 10^{-4}M$

$K_{sp} = [Ca^{2+}][F^-]^2$

$= (3.3 \times 10^{-4})(6.6 \times 10^{-4})^2$

$= 1.44 \times 10^{-10}$

22. $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$

x= solubility of AgCl in 1M $AgNO_3$

$$\underset{1M}{AgNO_3}(aq) \rightleftharpoons Ag^+(aq) + \underset{1M}{NO_3^-}(aq)$$

$[Ag^+] = x + 1 \simeq 1M \qquad (\because x << 1)$

$[Cl^-] = x$

$K_{sp} = [Ag^+][Cl^-]$

$1.8 \times 10^{-10} = (1)(x)$

$x = 1.8 \times 10^{-10}\,M$

23. $Ag_2CrO_4(s) \rightleftharpoons 2Ag^+_{(aq)} + CrO_4^{2-}(aq)$

$K_{sp} = [Ag^+]^2[CrO_4^{2-}]$

$= (5 \times 10^{-5})^2(4.4 \times 10^{-4})$

$= 1.1 \times 10^{-12}$

24. $Hg_2Cl_2 \rightleftharpoons \underset{s}{Hg_2^{2+}} + \underset{2s}{2Cl^-}$

$K_{sp} = [Hg_2^{2+}][Cl^-]^2$

$= (s)(2s)^2$

$K_{sp} = 4s^3$

25. $\underset{x}{Ag_2CrO_4} \rightleftharpoons \underset{2x}{2Ag^+} + \underset{x}{CrO_4^{2-}}$

x is the solubility

of Ag_2CrO_4 in 0.1M K_2CrO_4

$$\underset{0.1M}{K_2CrO_4} \rightleftharpoons \underset{0.2\ M}{2K^+} + \underset{0.1\ M}{CrO_4^{2-}}$$

$[Ag^+] = 2x$

$[CrO_4^{2-}] = (x + 0.1) \approx 0.1 \qquad \because x << 0.1$

$K_{sp} = [Ag^+]^2[CrO_4^{2-}]$

$1.1 \times 10^{-12} = (2x)^2(0.1)$

$1.1 \times 10^{-12} = 0.4x^2$

$$x^2 = \frac{1.1 \times 10^{-12}}{0.4}$$

$$x = \sqrt{\frac{1.1 \times 10^{-12}}{0.4}}$$

$$x = \sqrt{2.75 \times 10^{-12}}$$

$x = 1.65 \times 10^{-6}M$

26. When two are more solution are mixed, the resulting concentrations are differnet from the original.

Answers.indd 302 11-01-2021 17:36:21